

Bach Phan-Tat

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Data scientist with a background in finance and linguistics, currently a PhD candidate in computational linguistics. Skilled in quantitative analysis and machine learning, seeking opportunities to grow in AI and data science and apply data-driven methods to real-world problems.

WORK EXPERIENCE

Murf AI (speech technology company)

Nov 2024 – Present

Remote Research Consultant

Utah, USA

Project: Multilingual Text-to-Speech system

- Developed phonetic inventories and pronunciation lexicons (word–IPA mappings) for Vietnamese, Turkish, and multiple English varieties (US, UK, Scottish, and Australian), applying corpus-linguistic, phonological, and statistical methods to support multilingual TTS system development.
- Designed evaluation pipelines and test strategies for sentence segmentation models.
- Implemented Python scripts for automated text processing, lexical analysis, and language data preparation.
- Performed linguistic error analysis, documented findings, and recommended improvements for data and model quality.
- Reviewed relevant literature and industry practices to support solution design for multilingual speech technology tasks.
- Worked with cross-functional teams on annotation, transcription, and data preparation for speech and language workflows.

Vbee (speech technology and conversational AI company)

Feb 2023 – Aug 2024

Junior Research Engineer

Hanoi, Vietnam

Project: HUST Chatbot (a conversational AI for Hanoi University of Science and Technology)

- Developed and evaluated machine learning and BERT-based language models for classification tasks, including gender classification from names, chat domain classification, and toxicity detection, achieving accuracies above 85%.
- Designed statistical tests for data sampling and human evaluation.
- Developed a chatbot evaluation pipeline using commercial LLMs to assess response quality and system performance.
- Built and maintained language data workflows involving crawling, extraction, LLM-based data generation, pre-processing, annotation, and quality review for text and audio datasets.
- Contributed to the development of linguistic resources, prompts, and rule-based components for speech and language tasks, including speech-to-text, text-to-speech, and part-of-speech parsing.

what3words (location-mapping technology company)

Jul 2023 – Aug 2023

Remote Research Consultant

London, England

Project: Vietnamese Automatic Speech Recognition system (in collaboration with VinFast)

- Conducted linguistic error analysis of failed ASR outputs, identifying systematic patterns related to phonetic confusion and semantic ambiguity.
- Proposed data- and model-level improvements by combining linguistic analysis with machine learning insights.

ACADEMIC & PROFESSIONAL SERVICE

Supervising

- Advised a Master's student at Lancaster University on a Linguistics dissertation, which received a Distinction (76%). (2024).

Mentoring

- Mentored an applicant who was later accepted to the YSEALI Fellowship. (2024).
- Mentored an applicant who was later accepted to the PhD programme in Linguistics at Lancaster University. (2024).

Reviewing

- Reviewer for Linguists' Day, Linguistic Society of Belgium (LSB). (2025).
- Reviewer for Cognitive Linguistic Studies. (2026)

Organising

- Co-organised the PhD Day for the Department of Linguistics at KU Leuven. (2026).

PUBLICATIONS

Book chapters (edited volumes)

- **Bach Phan-Tat**^{*}, Sofia Aguilar-Valdez^{*}, Stefania Degaetano-Ortlieb, Dirk Geeraerts, & Dirk Speelman. (2025). Discursive parallels of the chemical revolution: topic modelling and distributional analysis. In *Large Language Models for the History, Philosophy, and Sociology of Science: Reflections from a Field in Motion* (open access, transcript). (Book chapter / edited volume). ^{*}Equal contribution.

Conference & workshop proceedings (peer-reviewed)

- **Bach Phan-Tat**, Kris Heylen, Dirk Geeraerts, Stefano De Pascale, Dirk Speelman (2026). Transparent Semantic Change Detection with Dependency-Based Profiles (LChange'26 Workshop @ EACL 2026).
- Khai Le-Duc, Tuyen Tran, **Bach Phan-Tat**, et al. (2025). MultiMed-ST: Large-scale Many-to-many Multilingual Medical Speech Translation. (EMNLP'25; Main Conference).
- Khai Le-Duc, Phuc Phan, Tan-Hanh Pham, **Bach Phan-Tat**, et al. (2025). MultiMed: Multilingual Medical Speech Recognition via Attention Encoder Decoder. (ACL 2025, Industry Track; Top 15% of accepted papers).
- Khai Le-Duc, Khai-Nguyen Nguyen, **Bach Phan-Tat**, et al. (2025). Sentiment Reasoning for Healthcare. (ACL 2025, Industry Track; Oral Presentation).

Pre-prints and Under-review articles

- **Bach Phan-Tat**, Kris Heylen, Dirk Geeraerts, Stefano De Pascale, Dirk Speelman (2026). Evaluating the Evaluator: Problems with SemEval-2020 Task 1 for Lexical Semantic Change Detection (Preprint - Under review)
- Catherine Wong, **Bach Phan-Tat**, Susan Fitzmaurice (2026). Methods, Data, and Conceptual Change: Reflections from Two Quantitative Diachronic Case Studies (Under review)
- **Bach Phan-Tat**, Kris Heylen, Dirk Geeraerts, Stefano De Pascale, Dirk Speelman (2026). ReFRAME or Remain: Unsupervised Lexical Semantic Change Detection with Frame Semantics (Preprint - Under review)
- Duc Cao-Dinh, Khai Le-Duc, Anh Dao, **Bach Phan-Tat**, et al. (2025). Audio-3DVG: Unified Audio - Point Cloud Fusion for 3D Visual Grounding. (Preprint).

PRESENTATIONS

Conference/workshop presentations

- **Bach Phan-Tat**, Kris Heylen, Dirk Geeraerts, Stefano De Pascale, & Dirk Speelman. (2026). Underlying Dimensions of Semantics Change. (Digital Humanities Congress 2026).
- **Bach Phan-Tat**, Kris Heylen, & Stefano De Pascale. (2026). Lexical Semantic Change Detection? A good parser might be enough! (LOT Conference)
- **Bach Phan-Tat**, Kris Heylen, Dirk Geeraerts, Stefano De Pascale, & Dirk Speelman. (2026). SynFlow: Continuous Semantics Change Analysis via Dependency Co-occurrences. (ICAME 47).
- **Bach Phan-Tat**, Dirk Geeraerts, & Dirk Speelman. (2026). Conceptual Change during the Chemical Revolution: Air, Acid and Water in the Royal Society Corpus. (ICAME 47).
- **Bach Phan-Tat**, Kris Heylen, Dirk Geeraerts, Stefano De Pascale, & Dirk Speelman. (2026). From Parsers to Prompts: Combining AI and Linguistics for Interpretable Language Evolution. (AI in Language Evolution Workshop @ Evolang 2026).
- **Bach Phan-Tat**. (2026). Slot-Filler Distributional Changes in Times of Scientific Debate: A case study of air. (Quantitative Diachronic Linguistics and Cultural Analytics: Data-Driven Insights into Language and Cultural Change).
- Khai Le-Duc, Khai-Nguyen Nguyen, **Bach Phan-Tat**, et al. (2024). Sentiment Reasoning for Healthcare. (AIM-FM Workshop @ NeurIPS 2024).

Invited talks & seminars

- **Bach Phan-Tat**. (2026). Underlying Dimensions of Semantics Change. (NLP & text mining seminar organised by Leuven.AI).
- **Bach Phan-Tat**. (2026). SynFlow: Continuous Semantics Change Analysis via Dependency Co-occurrences. (Data in Historical Linguistics seminar series).

EDUCATION

KU Leuven

PhD, Computational Linguistics

Thesis: **Underlying Dimensions in Conceptual Change**

Sep 2024 – Present

Leuven, Belgium

Selected Achievements

- Developed SynFlow, a Python package for analysing semantic change through syntactic co-occurrence patterns and Jensen–Shannon Divergence.
 - Implemented efficient syntactic pattern extraction and structuring using BFS/DFS, n-ary tree search, and rule-based filtering.
- Built automated linguistic annotation pipelines using local LLMs for corpus-based analysis.
- Cleaned, parsed, and lemmatised large diachronic and contemporary text corpora for downstream semantic analysis.
- Optimised computational workflows through parallel, batch, and distributed processing, reducing processing time by 80%.
- Maintained and extended NephoSem, a lexical semantics toolkit developed by the QLVL research unit.
 - Adapted the pipeline to support multiple word embedding types for downstream analysis.

Thesis supervisors: Prof. Dr. Dirk Speelman, Prof. Dr. Dirk Geeraerts, Dr. Kris Heylen

WorldQuant University

MSc, Financial Engineering

Finished modules: Financial Market, Financial Data

Oct 2025 – Present

Remote

University of Stirling
MSc, English Language and Linguistics

Sep 2021 – Sep 2022
Stirling, Scotland

Thesis: **The structure of *thời gian*: A semantic analysis of the Vietnamese lexeme *thời gian* based on Principled Polysemy**

Selected Achievements

- Randomly sampled, crawled, cleaned, and compiled a specialised corpus of 500 Vietnamese news articles using Python and #LancsBox.
- Identified seven senses of the Vietnamese lexeme *thời gian* ('time') and traced their semantic development using the Principled Polysemy framework.

Thesis supervisor: Dr. Andrew Smith

Thesis grade: 80% (Distinction; class rank: 1st)

Average grade: 78% (Distinction; class rank: 1st)

Academy of Finance

Aug 2016 – Aug 2020

BA, Corporate Finance

Hanoi, Vietnam

Relevant modules: Linear Algebra, Calculus, Probability and Statistics, Econometrics

TECHNICAL SKILLS

Programming & Computing

- Programming languages (ordered by proficiency): Python (NumPy, Pandas, Matplotlib, scikit-learn, spaCy, PyTorch, Hugging Face), R, Bash, LaTeX, MATLAB, Java
- Parallel, batch, and GPU-enabled computing workflows
- Databases: MongoDB, SQL
- Git/GitHub

Data Science

- Exploratory data analysis, data cleaning and (interactive) data visualisation
- Machine learning and deep learning for NLP, time-series, structured data and computer vision
- Quantitative research design, experimental design, and A/B testing
- LLM-based annotation and evaluation workflows
- Mathematics for data science: linear algebra, multivariate calculus, statistics

Linguistics & Corpus Methods

- Corpus tools: CQPweb, #LancsBox, Sketch Engine, AntConc
- Linguistic analysis: phonetics, morphology, syntax, semantics
- Linguistic annotation, transcription, and corpus preparation

AWARDS AND HONOURS

- 2026: 1st place, Databricks Challenge (USD 1,000); 4th Hack-Nation Hackathon.
- 2024: Marie Skłodowska-Curie Actions Doctoral Fellowship (3 years), Horizon Europe.
- 2022: Research-Based Learning Prize for the best Master's dissertation in Literature and Languages, University of Stirling (GBP 225); first student in the history of the Linguistics Department to receive the award.
- 2021: Postgraduate Scholarship, University of Stirling (GBP 4,000).

SELECTED ADDITIONAL TRAINING

Data Science

- Inferential Statistics (The University of Amsterdam)
- Applied Data Science Lab (WorldQuant University)
- Deep Learning (Neuromatch Academy)
- Applied AI Lab (WorldQuant University)

Linguistics

- Corpus Linguistics: Method, Analysis, Interpretation (Lancaster University)
- Corpus Linguistics Summer School (Birmingham University)

Financial Engineering

- Foundations of Financial Engineering (WorldQuant University)

Research Methodology

- Quantitative Method (The University of Amsterdam)
- Qualitative Research Methods (The University of Amsterdam)
- Open Science 101 (Neuromatch Academy x NASA)

LANGUAGES

- Vietnamese: Native
- English: IELTS 8.5 – C2
- Dutch: A2